

Changes in understory plant communities following soil scarification and amendments in a Northern Hardwood Forest

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ARTICLE INFO

Keywords:

American beech invasion
Functional diversity
Functional redundancy
Lime amendment
Sugar maple decline
Taxonomic diversity
Wood ash amendment

ABSTRACT

The regeneration dynamics of sugar maple-dominated stands in northeastern North America are intricately influenced by global changes such as soil acidification, invasive species, forest management, and climate shifts. A notable challenge is the expansion of American beech and its bark disease, threatening mature trees and ecosystem services. Although silvicultural practices aim to control beech regeneration and enhance high-value species, their impact on understory plant communities remains poorly documented. In Valcartier, eastern Canada, we conducted a field experiment to assess the short-term effects of soil scarification and diverse soil amendments (nitrogen fertilization, liming, wood ash) on understory plant communities in a beech-invaded sugar maple stand. Over four years, we inventoried vegetation and measured various biotic and abiotic variables. Results showed that soil scarification significantly reduced diversity metrics like Simpson dominance, functional diversity, and redundancy (a metric correlated to community stability and resilience to disturbance). Negative effects of scarification were exacerbated by nitrogen, lime, or wood ash additions. Lime amendment without scarification enhanced diversity, suggesting its potential to support understory community stability. While scarification and amendments can promote sugar maple, yellow birch, and pin cherry seedling establishment, they have collateral effects on the understory, impacting productivity, diversity, and resilience in the short term. Land managers should exercise caution when combining these treatments, as they may transiently alter the stability and resilience of the understory and other forest functions. A more sustainable approach involves applying scarification and amendments on small, well-distributed patches within the targeted stands for treatment, avoiding extensive application over large areas.

1. Introduction

Forest management practices aimed at soil amelioration have been implemented in northern hardwood forests with various objectives, including reducing soil acidity (Lundström et al., 2003; Reid and Watmough, 2014; Nolet et al., 2015), increasing calcium (Ca) concentration (Haynes and Naidu, 1998; Lundström et al., 2003), controlling American beech (*Fagus grandifolia* Ehrh) invasion, and improving sugar maple (*Acer saccharum* Marshall) reestablishment, growth and vigor (Kobe et al., 2002; Long et al., 2011). Although some of these practices, in addition to the removal of beech (Nyland et al., 2006) and the creation of large gaps by partial cutting (Poulson and Platt, 1996; Dracup and MacLean, 2018), have been shown to control beech invasion and promote sugar maple reestablishment (Nyland et al., 2006; Bognounou et al., 2023), the impacts of these disturbances on understory plants are not well understood.

The understory vegetation of northern hardwood forests plays a crucial role in shaping the ecological dynamics and biodiversity of these unique forest ecosystems. It serves multiple ecological functions by providing habitat and forage for a wide range of wildlife species, including birds, mammals, and insects (Gilliam, 2007; Landuyt et al., 2019). The diverse plant community in the understory layer offers shelter, nesting sites, and food resources, fostering biodiversity and promoting ecological interactions within the forest ecosystem. Furthermore, understory plants play a crucial role in numerous ecosystem services and functions by contributing to nutrient cycling, carbon sequestration, soil moisture retention, and erosion control, as well as to supporting soil health, and the overall forest productivity (Muller, 2014; Elliott et al., 2015).

Understanding the composition, dynamics, and ecological significance of the understory vegetation is essential for effective forest management and conservation strategies. Changes in the understory

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<https://doi.org/10.1016/j.foreco.2024.122089>

Received 18 November 2023; Received in revised form 4 June 2024; Accepted 12 June 2024

Available online 2 July 2024

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community can indicate shifts in forest health and successional stage, as well as reveal the impacts of disturbances, such as logging and invasive species (Franklin et al., 2000). Additionally, the understory serves as an indicator of environmental conditions, including light availability, moisture levels, and nutrient availability, providing valuable insights into the overall functioning of the forest ecosystem (Druckenbrod and Dale, 2012; Förster et al., 2017). Therefore, it is essential to understand the mechanisms by which forest management, including silvicultural and site preparation practices, influence the understory plant community assemblage to predict their potential impact on community stability and resilience.

Various methods have been proposed to predict the impact of disturbances on plant community assemblages, including taxonomic approaches (i.e. species richness, and diversity indices such as Shannon index) and species trait assemblages (Aubin et al., 2013; Pidgen and Mallik, 2013; Boiffin et al., 2015; Fourrier et al., 2015; Aubin et al., 2016). A universally accepted approach has not been established due to the complexity of ecosystems, the variability of disturbance events, and the stochasticity of community succession (Kayes et al., 2010; Pulsford et al., 2016). Nevertheless, the trait-based approach has recently demonstrated significant predictive success in modeling plant community assemblages (Keith et al., 2007; Muscarella et al., 2016; Ricotta et al., 2023a; Ricotta et al., 2023b). Besides enabling better quantification and prediction of disturbance effects on ecological communities (Mouillot et al., 2013), a growing body of evidence also indicates that functional traits drive environmental filtering and biological interactions, at least in part (Edwards et al., 2011; Venn et al., 2011; Edwards et al., 2013; Mastrogianni et al., 2021; Zheng et al., 2022).

The plant community assembly in post-disturbance environments is characterized by the progressive shift of the functional space, which is a multidimensional space that places species along functional traits. This shift is predicted to occur through two mechanisms: the niche and neutral theories (Webb et al., 2002). The neutral theory posits that stochastic fluctuations and dispersal filtering independently determine community assembly patterns (Tilman, 2004; Hubbell, 2005a, 2005b, 2006, 2008). Conversely, the niche theory emphasizes the role of environmental filtering (i.e., species with specific traits coexist under specific environmental stresses) and biological interactions (i.e., competitive exclusion causes limiting similarity) (Muscarella et al., 2016; Wang et al., 2022). According to the niche theory, functional traits associated with high resource exploitation ability dominate early succession and are gradually replaced by traits associated with the ability to exploit resources in a resource-poor environment (Grime, 1977). Similarly, under biological interactions, functional traits associated with strong colonization ability, such as shade intolerance, dominate early succession, but are replaced by traits associated with strong competition ability, such as shade tolerance.

Both the niche and neutral theories predict shifts in functional diversity, which refers to the distribution of species and their abundances in the functional space of a community (Mouillot et al., 2013). Specifically, functional diversity is expected to be low in early successional stages due to the dominance of species with high colonization and high resource exploitation ability, increase in intermediate succession stages due to the co-occurrence of early and late successional species, and decrease in late successional stages due to resource limitation and the dominance of species with high competitive ability (Muscarella et al., 2016). Functional redundancy (a metric positively correlated to community stability and resilience to disturbance (Biggs et al., 2020), which refers to the number of species having the same ecosystem roles (Mouillot et al., 2013), may follow a similar pattern as functional diversity. However, different disturbance types can lead to alternative scenarios (Pulsford et al., 2016).

Different disturbance types have been tested in northern hardwood forests to reduce the dominance of beech over other hardwood species such as sugar maple. These disturbances include the removal of beech trees, saplings, and seedlings (Nyland et al., 2006; Moore et al., 2012;

Nolet et al., 2015; Bédard et al., 2022), the creation of large gaps with partial cuts to increase light availability (Poulson and Platt, 1996; Duchesne et al., 2013; Nolet et al., 2015; Dracup and MacLean, 2018; Bédard et al., 2022), the use of amendments to correct soil acidity and enhance soil fertility (Poulson and Platt, 1996; Moore et al., 2012; Duchesne et al., 2013; Reid and Watmough, 2014; Nolet et al., 2015; Dracup and MacLean, 2018), and scarification to expose the mineral soil to facilitate seed germination, seedling survival and growth (Gastaldello et al., 2007b; Natural Resources Canada, 2019; Bédard et al., 2022; Bognounou et al., 2023). While some treatments, such as selective cut and soil scarification, have been studied, the combined effect of soil scarification and amendment on the understory plant community assembly in the northern hardwood forests remains unknown. Selection cut associated with soil scarification promotes both asexual and sexual reproduction modes and shade-intolerant species in northern hardwood forests (Newmaster et al., 2007; Jaeger et al., 2022), whereas soil amendment may promote species with high fertility requirements. Lime and wood ash amendments, for instance, increase Ca concentration and pH, improve soil structure and hydraulic conductivity (Haynes and Naidu, 1998; Lundström et al., 2003), and increase earthworm activity and microporosity (Bolan et al., 2003). Additionally, nitrogen (N) fertilizer increases N availability (Marschner, 2012) and breaks the seed dormancy of some species (Auchmoody, 1979). Therefore, investigating the combined impact of soil scarification and amendment on the understory plant community assembly is crucial for understanding the mechanisms underlying their effects on community stability and resilience in the early successional stage of northern hardwood forests.

We conducted a field experiment in Valcartier, eastern Canada, to evaluate the short-term impact (4 years) of various soil amendments and soil scarification techniques on understory plant community assemblages in a sugar maple-dominated stand under beech invasion. Our findings shed light on the potential collateral effects of these management practices on the assembly of understory plant communities. Our research questions are:

1. To what extent do soil scarification and amendment induce functional changes in the structure of the understory plant community? We expected that soil scarification and amendment would induce significant changes in the diversity structure of the understory plant community with scarification reducing functional diversity through homogenization of the soil structure.
2. Among the environmental drivers of understory plant communities' functional structure, how significant are the influences of soil scarification and amendment? We expected that the association of soil scarification and amendment would be the most influential drivers of the diversity structure of the understory plant community due to the alteration of both the soil physical structure and soil chemistry.
3. Do specific trait assemblages of understory plant communities show associations with different soil treatments, such as scarification and amendment? We expected that soil treatments would significantly influence the understory plant community, leading to distinct trait assemblages based on functional groups and species richness. Scarified soils would show a higher abundance of species with traits such as rapid growth, high seed dispersal capacity, and stress tolerance, due to the increased soil disturbance and resource availability. Amended soils would demonstrate an increase in species with traits associated with nutrient acquisition, due to the enhanced soil nutrient levels. Scarified and amended soils would display a unique assemblage of traits, reflecting the combined effects of increased disturbance and nutrient availability, potentially leading to a more diverse and competitive plant community.

2. Material and methods

2.1. Study location

The study was conducted in the Boreal Shield ecozone of Eastern Canada at the Canadian Forces Base of Valcartier, situated near Quebec City at coordinates 46.91° N, 71.58° W (refer to Fig. 1). The site experiences a mean minimum temperature of -18.8°C in January and a mean maximum temperature of 23.8°C in July, with an average annual precipitation of 1460 mm (Canadian Climate Normals 1981–2010 Station Data – Rivière Verte Ouest). The trial was established in a mature sugar maple-dominated stand that was approximately 70 years old and covered an area of around 24 ha. The stand was severely affected by beech-bark disease and exhibited clear signs of sugar maple regeneration failure, with a significant portion of ground cover devoid of regeneration (33 %), few sugar maple seedlings (<20 %), and a high density of beech saplings (50 %). In 2014, a selection cut was performed, which removed most of the beech trees and retained 65 % of the initial forest cover. The stand developed on sandy loam till material that originated from granitic/gneiss formations of the Canadian Shield. The soil is classified as Dystric Brunisols, which belong to the Inceptisols order in the US soil taxonomy and the Cambisols order in the World Reference Base (WRB, 2022). The site exhibited acidic conditions (soil pH = 4.5) at the beginning of the experiment (Bognounou et al., 2023). More information on site characteristics including soil data can be found in Bognounou et al. (2023).

2.2. Experimental design

In the fall of 2017, we established a randomized blocks field trial to evaluate the effects of two levels of soil scarification (control and scarification) and four levels of amendments (control, lime, ash, and fertilizer) on species re-establishment. We started by clearing the understory

cover, which mainly consisted of beech saplings, and removing all woody debris larger than 1 cm in diameter. We then laid out 12 plots, each measuring $8 \times 4 \text{ m}$ and containing four paired subplots of $2 \times 2 \text{ m}$. To ensure randomization, we assigned soil scarification to one subplot in each column and assigned amendment treatments (lime, wood ash, fertilizer, and control) to each column. The control amendment consisted of a subplot without any amendment. Each plot comprised eight subplots, with four scarified, four non-scarified, two amended with lime, two with ash, two with fertilizer, and two with no amendment (Fig. 1 & S1).

To assess the effect of scarification on species re-establishment, we lightly scarified four subplots in each plot in May 2018 using a toothed disk (UGS: 07 TS) mounted on a brush cutter (STIHL FS 361 C-EM). This method disturbed a shallow depth of the soil surface (1–2 cm), mixed the organic layer with the mineral soil, and partially exposed the mineral soil. To evaluate the effect of each amendment without scarification on soil amelioration and species re-establishment, we applied lime (0.5 kg/m^2 of CaCO_3), wood ash (0.885 kg/m^2), and fertilizer (0.5 kg/m^2 of NH_4NO_3) to one non-scarified subplot within each block. We also applied lime, wood ash, and fertilizer to one scarified subplot within each plot to evaluate the effect of their interaction with soil scarification on species re-establishment. Finally, we used one scarified and non-scarified subplot that did not receive any amendment within each plot to assess the effect of soil scarification without amendment.

The ash used in the trial was bottom ash collected from a nearby industrial biomass boiler (Resolute Forest Products, Château-Richer), which mainly used conifer bark as feedstock (see Table S1 for ash chemical properties). The lime (powdered limestone, Mosher Limestone, Upper Musquodoboit, NS) and fertilizer (ammonium nitrate $\geq 95\%$ ACS, Thermo Scientific Chemicals) were purchased from certified soil amendment providers.

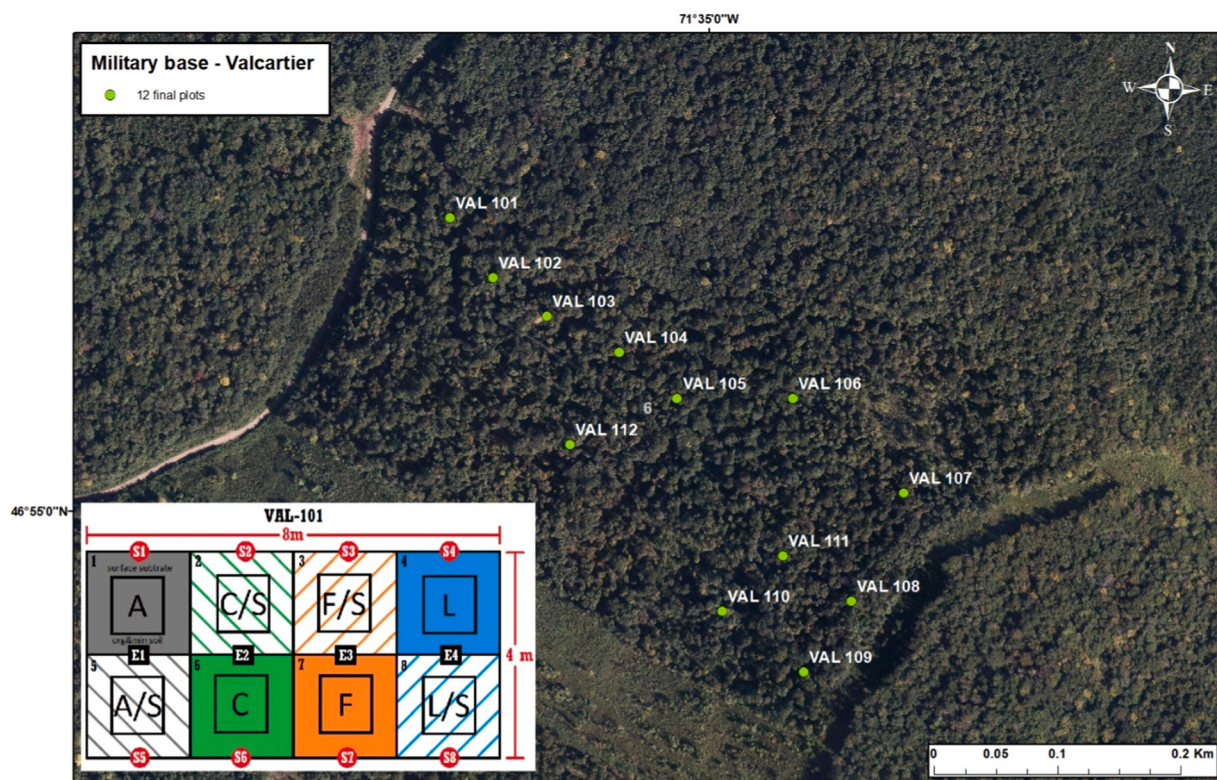


Fig. 1. Map of the study site showing the location of plots within the sugar maple-dominated stand in Valcartier. An example of a plot layout is represented in the lower-left corner (A = Ash; C = Control; F = Fertilizer; L = Lime; S = Scarification). Plots consisted of 8 subplots where the different treatments were applied. Each subplot included a buffer strip of 0.5 m around the measurement area.

2.3. Data collection

We collected data on biotic and abiotic variables during the study season and year (September 2017 to October 2021). Biotic variables refer to the abundance of woody plant species, while abiotic variables refer to soil properties and light availability. Prior to site preparation, we counted the number of woody species individuals and classified them by species and growth stage, which were coded as follows: Seedling as species name_Reg (e.g., *A.saccharum_Reg*); sapling as species name_shrub (e.g., *A.saccharum_shrub*); and tree as species name_Tree (e.g., *A.saccharum_Tree*). Soil cores (0–20 cm) were collected from the margin of each subplot to determine standard soil properties. We determined soil texture (sand, silt, and clay) following the hydrometer method (Kroetsch and Wang, 2007) and recorded surface substrate type (organic, mineral, buried wood, stony) following soil core removal.

The monitoring of treatment effects on understory vegetation diversity began after the application of treatments in 2018. We conducted the inventory of understory vegetation in summer and fall, except in 2020 when the inventory was conducted only once, in summer, due to Covid-19 related restrictions. During each inventory, we identified and recorded each understory species and estimated its cover in each subplot.

To assess the light availability in the understory strata, we conducted photosynthetic photon flux density (PPFD) measurements 4–5 times during the growing season using Li-200R pyranometers (Li-Cor, Nebraska, USA). The measurements were taken at the center of each subplot, while continuous measurements were performed at an adjacent opening to record the above canopy PPFD using a Li-1500 datalogger. We calculated the light availability for each subplot as %PPFD. Soil moisture measurements were taken at the same time as the light availability measurements using a Fieldscout TDR 150 soil moisture meter.

2.4. Selection of traits and ecological performances

We selected ten plant species' functional traits that are linked to various ecosystem functions (see Table 1). The selected traits reflect how plants adapt to forest disturbances, including their survival (e.g., Raunkiaer's growth form), vegetative reproduction (Gastaldello et al., 2007a), colonization (e.g., seed dispersal modes, seed persistence, seed dormancy; (Natural Resources Canada, 2019), and growth (e.g., shade tolerance, moisture preference) capacities. To obtain the traits for all species, we extracted them from online plant databases (Traits of Plants in Canada database (TOPIC): <http://www.nrcan.gc.ca/forests/research-centres/glf/20303> and United States Department of Agriculture (USDA) plant database: <https://plants.usda.gov>, <https://www.feis-crs.org/feis>).

2.5. Data analysis

All analyses were conducted using R statistical and programming environment version 4.1.3 (R Core Team, 2022). The data analysis followed a protocol for data exploration as described by Zuur et al. (2010). This included checking for outliers in the dependent and explanatory variables, assessing homogeneity, normality, and zero-inflation of the dependent variables, examining collinearity between explanatory variables, exploring relationships between dependent and explanatory variables, and assessing interactions and independence of the dependent variables. We used the “fitdistrplus” package (Delignette-Muller and Dutang, 2015) to explore dependent variables, which revealed that none had a normal distribution. We also used variance inflation factors (VIF) to investigate collinearity between explanatory variables, and found that none of the variables had a VIF greater than 2.5, indicating no collinearity among the explanatory variables (Zuur et al., 2007). The statistical analyses were carried out in multiple steps.

Table 1

Details on the selected functional traits (for additional information, see (Cornelissen et al., 2003; Laliberté and Legendre, 2010)).

Metric	Categories	Link with response to environment	Link with effects on ecosystem function
Life history			
Raunkiaer's life form	Phanerophyte (buds and vegetative parts above the soil surface, such as trees and shrubs) Chamaephyte (buds and vegetative parts close to the ground, such as herbs and subshrubs) Hemicryptophyte (buds and vegetative parts at or just below the soil surface, such as bulbs and tubers) Geophyte (plants with perennating organs below the soil surface, such as bulbs, tubers, and rhizomes) Non-vascular plant species (not part of the Raunkiaer's life form classification per se)	Bud placement relative to forest floor affects the ability to survive disturbance (re-sprouting ability)	Climate regulation (e.g., carbon sequestration in vegetation and soil), soil fertility, and nutrient cycling (higher nutrient uptake with increased diversity in terms of life form)
Use of resources			
Plant height	Continuous (cm)	Potential for competitiveness	Climate regulation (e.g., carbon sequestration in vegetation and soil)
Moisture preference	Hydric Hydric-Mesic Mesic Mesic-xeric Xeric	Potential for colonization and competitiveness	Water regulation, soil fertility and nutrient cycling
Light requirement	Shade intolerant Mid tolerant Shade tolerant	Potential for colonization and competitiveness	Soil fertility and nutrient cycling
Nutrients requirement	High Medium Low	Potential for colonization and competitiveness	Soil fertility and nutrient cycling
Mobility			
Dispersal modes	Autochory Anemochory Zoochory	Potential for invasion after disturbance	Provision of trophic resources (e.g., to frugivore and granivore)
Reproduction			
Vegetative propagation	Non clonal Suckering Sprouting	Potential for recolonization from surviving individuals, or regrowth following top kill	Climate regulation, provision of trophic resources (e.g., herbivores)
Seed production	Abundant Semi-abundant Scarce	Potential for invasion after disturbance	Provision of trophic resources
Seed dormancy	Dormant Non dormant	Potential for reestablishment after disturbance	Soil fertility and nutrient cycling
Seed persistence	Transient-absent Semi-persistent Persistent	Potential for reestablishment after disturbance	Soil fertility and nutrient cycling

2.6. Analysis of soil scarification and amendment effects on the functional changes in the understory plant community

To evaluate the impact of soil scarification and amendment on understory plant community functional structure, we used the species dominance, functional redundancy, and functional diversity (DRQ) ternary diagram (Ricotta et al., 2023) to express the relationship among D (Simpson dominance), R (Functional redundancy) and Q (Functional diversity). The Simpson dominance (D) measures diversity by considering the relative abundance of each species, the functional redundancy (R) assesses the number of species performing similar roles in an ecosystem (De Bello et al., 2007), and the functional diversity (Q) gauges the range of ecological functions represented in a community based on trait divergence between species.

To assess functional diversity, we calculated Rao's quadratic diversity Q (Rao, 1982), which integrates species richness, evenness, and divergence components (Mouchet et al., 2010). Q is the expected dissimilarity between two individuals chosen at random with replacement from the subplot:

$$Q = \sum_{i,j=1}^N p_i p_j d_{ij}$$

Where N is the number of species, p_i and p_j are the proportion of individuals to the i and j -th species in the community, d_{ij} is pairwise dissimilarities between species i and j . The functional redundancy (R) was estimated by comparing taxonomic and functional diversity indices (R) based on their functional dissimilarity (De Bello et al., 2007):

$$R = S - Q$$

Where Q is Rao's quadratic diversity and S is the Simpson diversity (Simpson, 1949):

$$S = 1 - \sum_{i=1}^N p_i^2$$

The species dominance D was calculated as the probability that two individuals selected at random with replacement from a given plot belong to the same species:

$$D = 1 - S, \text{ where } S \text{ is the Simpson diversity.}$$

We used the *rao.diversity* function within the "SYNCSA" package version 1.3.4 (Debastiani and Pillar, 2012) to compute S, Q and R. To test the effect of the combined effect of soil scarification and amendment over year and season on understory diversity structure, we ran Permutational Multivariate Analysis of Variance (999 permutations) using distance matrices partitioning with D, R, and Q as response variables, and soil scarification, soil amendment, year and season (Year_Season) and their second and third interaction terms as explanatory variable. We used the *anodis2* function within the "vegan" package version 2.6–4 (Oksanen et al., 2013).

2.7. Classification of the environmental drivers of understory plant communities' functional structure

To assess the influence of microsite conditions on the understory plant community, we conducted a Random Forest (RF) analysis to identify the most significant predictors of the six diversity indices (Simpson Dominance, Functional diversity, Functional redundancy, Simpson Diversity, Functional uniqueness and Functional homogeneity) from a set of microsite variables. This analysis involved modeling subsamples of observations and variables (Breiman, 2003). We included biotic and abiotic predictor variables in our analysis. The biotic predictor variables consisted of initial woody plant abundance (*B.alleganiensis*0, *B.papyrifera*0, *A.spicatum*0, *A.pensylvanicum*0, *A.rubrum*0, *A.saccharum*0, *F.grandifolia*0, *A.balsamea*0, *B.alleganiensis* Shrub, *A.pensylvanicum* Shrub, *A.rubrum* Shrub, *A.saccharum* Shrub, Deciduous sp_Shrub of all other species not previously mentioned, *F.*

grandifolia Shrub, *A.pensylvanicum* Tree, *A.balsamea* Tree), regeneration or specific seedling abundance during the experiment (*B.alleganiensis*, *B.papyrifera*, *A.spicatum*, *A.pensylvanicum*, *A.rubrum*, *A.saccharum*, *F.grandifolia*, *A.balsamea*, *V.alnifolium*). The biotic variables can significantly influence species establishment by inducing regeneration via sprout, sucker, and seed rain (initial wood plant abundance) or by affecting resource availability (e.g., nutrients, light, water) and plant interactions (e.g., competition, facilitation, allelopathy, pathogens).

The abiotic predictor variables include pre-treatment soil texture (sand, clay, silt), soil class (sandy loam, loam), soil substrate (buried_wood, organic, stony, mineral), post-treatment soil properties (Soil_moisture_2018, Soil_moisture_2019, Soil_moisture_2020), and light availability (Light_2018, Light_2019, Light_2020). We also included the time since treatment (Year_Season) as a predictor.

We used a tree classification algorithm to build numerous trees, based on randomly sampling the predictors and the observations. The prediction model is formed by the average prediction of all the regression trees, forming the "random forest". Each tree is grown using the classification tree methodology without pruning. We used the "randomforest" package in R (Liaw and Wiener, 2002) to quantify the importance of independent variables. The default value of "mtry" (the number of variables available for splitting at each tree node, here equal to the square root of the number of predictor variables) was used, given conflicting evidence in the literature about the influence of this variable (Cutler et al., 2007; Strobl et al., 2008). However, the importance ranking may vary from one run to another due to the random selection of training data and variables used to determine the split at each node in Random Forest (Millard and Richardson, 2015). Therefore, we ran the model with 1000 trees to deal with the variation of the importance value (Behnamian et al., 2017).

We used the mean increase in accuracy (%IncMSE) to measure predictor influence. %IncMSE is the average increase in squared residuals of the test set when the variable is permuted. The higher the value, the higher the variable contribution to the overall variance of the predicted variable. This performance indicator is drawn separately for each tree of the forest and finally averaged over all trees of the forest and finally averaged over all trees. In our analysis, we preferred %IncMSE over the indicator on the increase node purity (IncNodePurity) based on the Gini Index because it is the most robust and informative measure, while IncNodePurity is biased and should only be used if the extra computation time of calculating %IncMSE is unacceptable (Welling, 2017).

2.8. Analysis of trait assemblages' associations with microsite conditions

We used three matrices for this analysis: (1) L, which represented community data with species as columns and subplots as rows; (2) Q, which represented trait data with traits as columns and species as rows, and (3) R, which represented environmental data with microsite conditions (i.e., the most influential drivers from RF: Year, Season, soil moisture, light availability, sand, clay, silt, *F.grandifolia* Shrub, *F.grandifolia*0, and *A.pensylvanicum* Shrub) as columns and subplots as rows. To evaluate the global significance relation between traits and microsite conditions, we performed an RLQ analysis using the *rlq* function of the "ade4" package (Chessel et al., 2004; Dray and Dufour, 2007; Dray et al., 2007; Bougeard and Dray, 2018; Thioulouse et al., 2018). The preliminary step of RLQ analysis involved separate multivariate analyses on each table. We applied correspondence analysis to the community data and Hill-Smith multivariate analysis (Hill and Smith, 1976) to the traits and environmental data, which contained both quantitative and categorical variables. The separate analyses of traits and microsite conditions were weighted by the species weight derived from the previous correspondence analysis. The results of these multivariate analyses were then used as input into the RLQ analysis, which performed a double inertia analysis of two matrices (R and Q) with a link expressed by a community matrix (L) (Dray and Legendre, 2008).

To test the significance of the associations between individual traits

(Q) and microsite conditions (R), we conducted a fourth-corner analysis using the *fourthcorner* function of the "ade4" package. We used a model 6 type and Monte Carlo permutation test with 9999 permutations (Legendre et al., 1997; Peres-Neto et al., 2017). The permutation "model 6" combines permutation "model 2" (i.e., permute entire rows of table L) and "model 4" (i.e., permute entire columns of table L) to test the null hypothesis that microsite conditions do not influence the distribution of species with fixed traits and that species characteristics do not influence the species composition of the subplots.

We also calculated the community weighted mean (CWM) of functional traits to provide additional insight into trait assembly. The CWM is the average trait values (or proportion for categorical traits) for understory plants at each subplot level weighted by species cover estimation. To calculate the CWM, we used the *dbFD* function of the "FD" package (Laliberté and Legendre, 2010; Laliberté et al., 2014). We run Permutational Multivariate Analysis of Variance (999 permutations) using distance matrices partitioning with CWM of functional traits as response variables, and soil scarification, soil amendment, Year, Season and their second and third interaction terms as explanatory variable.

3. Results

3.1. Understory plant community diversity after soil scarification and amendment

The test of soil scarification and amendment effects on understory diversity structure (specifically DRQ composition) using permutation distance-based multivariate analysis of variance revealed a significant interaction between soil scarification and amendment ($p = 0.031$; Table 2 and Fig. 2A), and a significant effect of the main terms Year ($p = 0.038$; Table 2 and Fig. 2B), and Season ($p = 0.001$; Table 2 and Fig. 2C). Specifically, soil scarification led to a significant decrease in taxonomic diversity, functional diversity, and functional homogeneity, and an increase in functional uniqueness and the dominance of a few species. Moreover, this impact was intensified when fertilizer, ash, and lime were added (Fig. 2A). For instance, *Viburnum alnifolium* was only the dominant species on scarified soil once it was amended with lime (Figure S2). Moreover, lime amendment to non-scarified soil resulted in significantly higher levels of taxonomic diversity, functional diversity, functional redundancy, functional homogeneity, and lower level of functional uniqueness i.e., the codominance of many species (e.g., *Ribes glandulosum*, *Medeola virginiana*, *Dryopteris marginalis*, *Dennstaedtia punctilobula*) among the understory vegetation compared to the other treatments.

Additionally, from 2018 to 2019, there was a significant increase in taxonomic diversity, functional diversity, and functional homogeneity, followed by a decrease in 2020 and 2021, resulting in an intermediate state (Fig. 2B). There was also a significant seasonal decrease in

taxonomic diversity, functional diversity, and functional homogeneity, and conversely seasonal increase in functional uniqueness, from early summer to fall (Fig. 2C).

3.2. Drivers of understory plant community diversity and resilience

The random forest (RF) analysis yielded significant insights into the drivers of understory vegetation diversity structure, with microsite conditions explaining about 40 % of the observed variances (Table 3). Notably, the importance of microsite drivers on understory vegetation diversity varied across different indices (Fig. 3).

Among all metrics, the cumulative effect of soil scarification and amendment emerged as the most influential driver, followed closely by abiotic factors such as light availability and soil moisture, and biotic factors such as beech shrub and seedling abundance. Soil amendment without scarification exerted a greater impact on taxonomic diversity (Fig. 3 A&D) as well as on functional diversity and homogeneity (Fig. 3 B&F) relative to scarification without amendment. However, the reverse was true for both functional redundancy and uniqueness (Fig. 3 C&E). From 2018–2020, we observed a significant shift in the factors influencing all diversity indices. Specifically, light availability emerged as an increasingly critical determinant of understory vegetation structure over the study period. Conversely, the importance of soil moisture as a driver of understory diversity showed a marked decline during the same timeframe.

Beech shrub and seedling abundance before and after site preparation emerged as additional key drivers significantly impacting all diversity indices. Furthermore, seasonality played a more substantial role than year in influencing functional redundancy and uniqueness. On the other hand, the importance of other microsite drivers appeared relatively lower, with biotic factors generally ranking below abiotic factors in significance.

3.3. Variation of functional trait assembly after soil scarification and amendment

The exploration of the relationship between the understory functional traits (Figure S3A) and microsite variables (Figure S3B) showed that the first two axes of the RLQ analysis captured 72 % of the total association between functional traits and microsite conditions (Table 4, $P < 0.001$, Monte-Carlo 2000 permutations "model2" test). The first RLQ axis (37 % variation explained) highlighted the differentiation of functional traits between scarified and non-scarified soils. The second axis (35 % variation explained) revealed a distinct separation of functional traits based on soil properties. Furthermore, the RLQ analysis revealed that soil treatments, particularly scarified soil with both ash and lime amendments and non-scarified soil with fertilizer amendment, had stronger associations (indicated by longer arrows) with functional traits compared to other environmental variables.

According to the fourth-corner analysis, the synergistic effects of scarification and lime amendment were found to promote hemi-cryptophyte, mesic-xeric, and semi-persistent species (Figure S4). Scarification, when coupled with lime, benefited for shade-tolerant species, albeit negatively impacting mesic species. In contrast, lime application without scarification favored species with medium fertility requirements but negatively affected species with low fertility requirements. The introduction of ash in scarified soil had a positive impact on xeric species and those with non-dormant seeds, while concurrently exerting a negative effect on species with dormant seeds. Similarly, the positive influence of ash on species with scarce seeds was observed without scarification. On the other hand, scarification combined with fertilizer application favored species with scarce seeds but had a negative effect on hydric species. Fertilizer application without scarification, however, promoted species with medium fertility requirements, mesic-xeric species, and autochore species, while negatively affecting mid-tolerant species.

Table 2

Permutational Multivariate Analysis of Variance results testing the effect of soil treatments over Seasons and Years on understory diversity structure (DRQ) using distance matrices partitioning (Df= Degree of freedom; SumOfSqs= Sum Of Squares; R^2 = Coefficient of determination; F= F-statistics; $\Pr(>F)$ = p-value associated with the F-statistic).

Variable	Df	SumOfSqs	R^2	F	$\Pr(>F)$
Amendment	3	1.175	0.035	7.379	0.001
Scarification	1	1.240	0.037	22.364	0.001
Year	3	0.437	0.013	2.741	0.038
Season	1	0.968	0.029	18.243	0.001
Amendment:Scarification	3	0.489	0.015	3.070	0.023
Amendment: Year	9	0.565	0.017	1.182	0.299
Scarification: Year	3	0.377	0.011	2.368	0.063
Year:Season	1	0.003	0.000	0.052	0.842
Amendment:Season	3	0.120	0.004	0.755	0.505
Scarification:Season	1	0.097	0.003	1.830	0.182

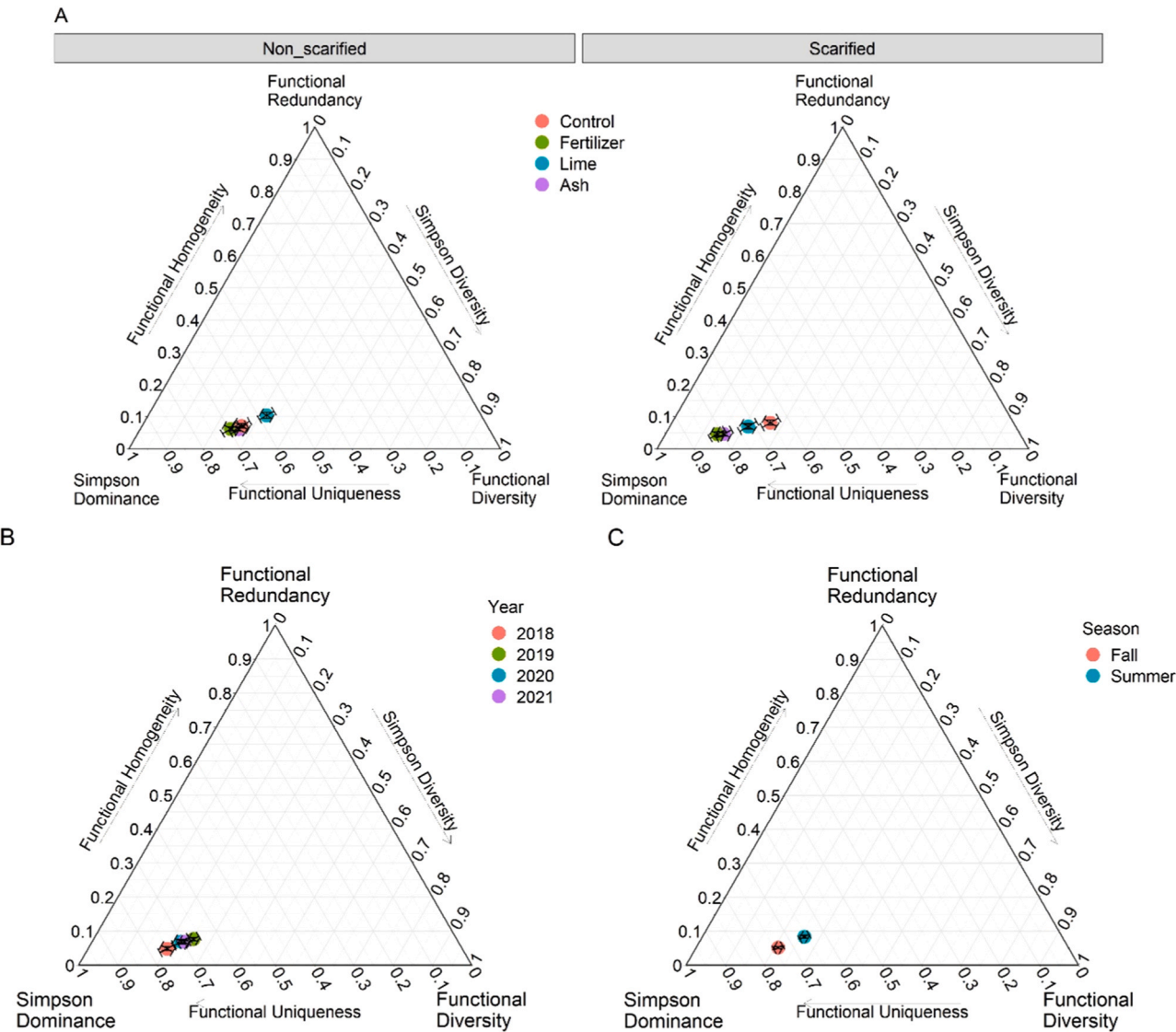


Fig. 2. Ternary diagram illustrating the interplay between species dominance, functional redundancy, and functional diversity in scarified and non-scarified soils across various amendments (A), during the experiment (B) and between Seasons (C). In this ternary diagram, the vertices of the triangle correspond to the three components and each corner of the triangle refers to a situation where the value of one component equals 1, and the other two values are zero. The contrast between a vertex to the opposite edge of the triangle reflects three different facets of the functional structure of the understory plant community: Simpson dominance (D) versus Simpson diversity ($R + Q$); Functional diversity (Q) versus Functional homogeneity ($D + R$); Functional redundancy versus Functional uniqueness ($D + Q$).

Table 3
Summary of the Random forest analysis identifying the most significant abiotic and biotic predictors of the six diversity indices.

Indices	Mean Squared Residuals	Percentage of variance explained
Species dominance	0.036	39.43
Functional diversity	0.022	40.58
Functional redundancy	0.003	35.70
Simpson diversity	0.036	39.32
Functional uniqueness	0.003	35.30
Functional homogeneity	0.022	40.63

The analysis of community-weighted mean traits (CWM) revealed a distinct assembly of dominant traits within the community, characterized by preferences for high nutrient levels, moderate shade tolerance, mesic moisture conditions, as well as traits associated with geophytes, prolific seed production, and transient seed strategies (Table 5 and Fig. 4). The application of different soil amendments significantly

enhanced the effects of scarification, leading to an increased presence of traits indicative of both high and medium nutrient demands, shade intolerance, mesic moisture preferences, and life forms such as phanerophytes, hemicryptophytes, non-vascular plants, and species with persistent seeds. Conversely, the application of amendments in conjunction with scarification was observed to mitigate traits associated with lower nutrient needs, high shade tolerance, preference for hydric conditions, and the chamaephyte life form.

4. Discussion

4.1. Effect of soil scarification and amendment on understory plant community

Soil scarification reduced the diversity structure (i.e., Simpson dominance, functional diversity and functional redundancy) of understory plant communities, and this effect was further amplified with either the addition of nitrogen fertilizer, lime, or wood ash. This finding suggests that soil scarification associated with amendment decreases the

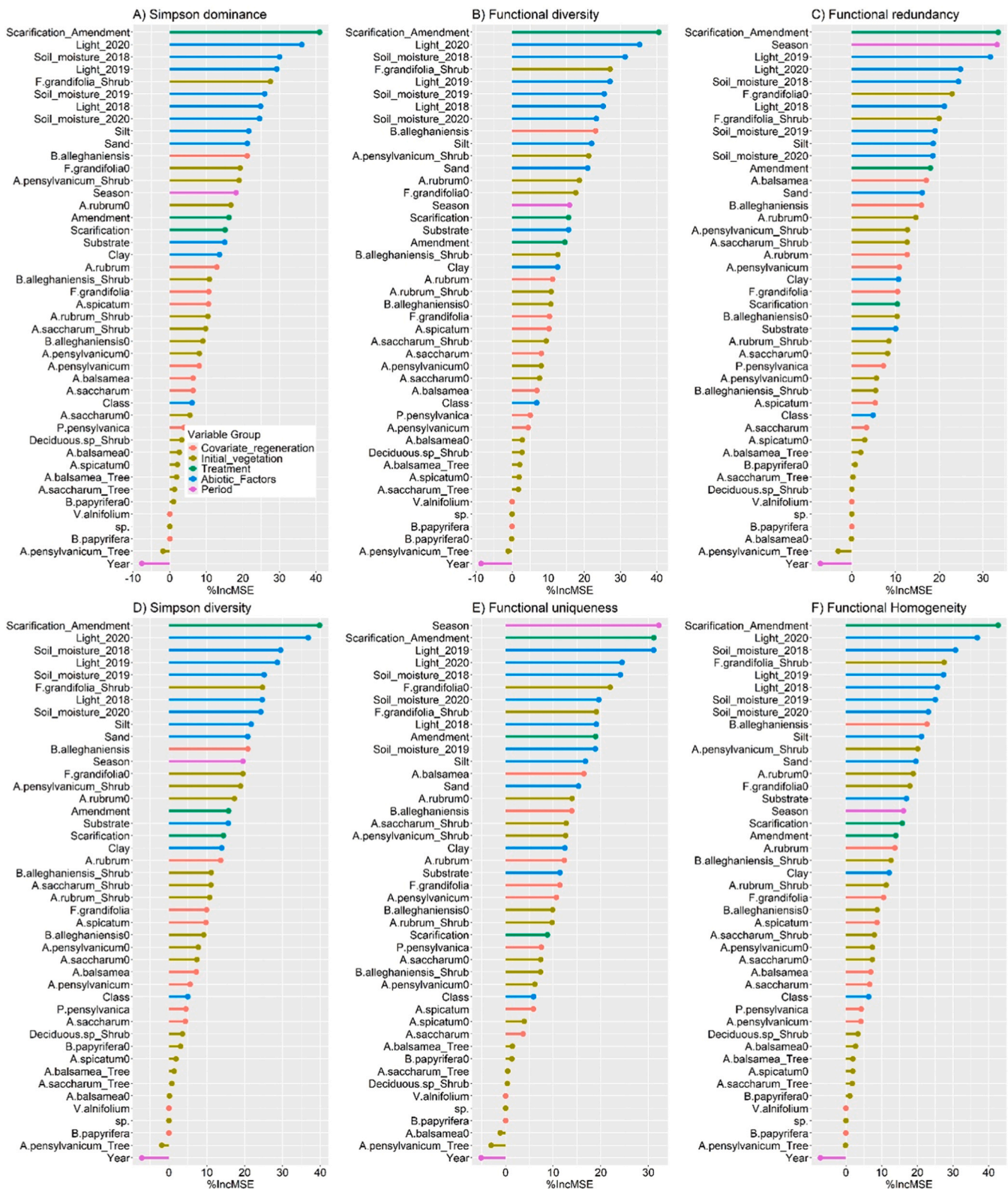


Fig. 3. Random forest analysis showing the influence of abiotic and biotic factors on understory vegetation diversity structure. The percentage (%) increases in Mean Squared Error (MSE) after a driver was randomly permuted.

stability and the resilience of understory plant community in the early stage of recovery (4 years). Alternatively, lime amendment without scarification increased the diversity structure of understory plant communities suggesting that lime amendment without scarification increases their stability and resilience. These findings are consistent with

our hypothesis and in line with previous research that reported that soil disturbance may constrain understory species dispersal and recruitment processes (Aschehoug et al., 2016; Raymond et al., 2018; Jaeger et al., 2022; Lanta et al., 2023). Scarification, which involves removing the top layer of soil and vegetation, creates a disturbance that alters light

Table 4

Summary of the RLQ analysis: eigenvalues and percentage of total co-inertia accounted for by the first two RLQ axes, covariance refers to the covariance between the two new sets of factorial scores projected onto the first three RLQ axes (square root of eigenvalue); correlation refers to the correlation between the two new sets of factorial scores projected onto the first three RLQ axes; cumulative inertia refers to the variance of each set of factorial scores computed in the RLQ analysis, both for the environment and for the traits.

Variables	Axis 1	Axis 2
Eigenvalues	1.77	1.69
% of total co-inertia	37.00	35.33
Covariance	1.33	1.30
Correlation	0.48	0.40
Cumulative inertia (microsite)	2.79	5.17
Cumulative inertia (traits)	2.79	7.13

Table 5

Permutational Multivariate Analysis of Variance results testing the effect of soil treatments and Year_Season on understory community weighted mean (CWM) using distance matrices partitioning (Df = Degree of freedom; SumOfSqs = Sum Of Squares; R² = Coefficient of determination; F = F-statistics; Pr(>F) = p-value associated with the F-statistic).

Variable	Df	SumOfSqs	R ²	F	Pr(>F)
Amendment	3	1.169	0.027	5.417	0.001
Scarification	1	1.008	0.025	14.007	0.001
Year	3	0.367	0.009	1.698	0.116
Season	1	0.075	0.002	1.041	0.347
Amendment:Scarification	3	0.136	0.027	5.261	0.001
Amendment: Year	9	0.180	0.004	0.278	1
Scarification: Year	3	0.308	0.007	1.429	0.184
Year:Season	1	0.036	0.001	0.498	0.662
Amendment:Season	3	0.067	0.002	0.312	0.957
Scarification:Season	1	0.172	0.004	2.395	0.092

availability, soil properties and the plant community structure. This may result in changes in species composition, diversity, and ecosystem functioning (Ratcliffe et al., 2017). Though scarification can improve seed germination and soil amendment can improve fertility and soil structure, they can also have negative impacts on understory species richness (Tullus et al., 2019). For instance, scarification and amendment can create new opportunities for species with specific traits (suckering or sprouting ability, invasive species), alter soil chemistry and nutrient availability, potentially reducing diversity structure. Scarification can disrupt the soil seed bank, which reduces the diversity of understory plants that emerge in the following seasons. Scarification also damages or removes existing vegetation. Soil amendment, on the other hand, has short-term positive effects on the understory plant community diversity (Thomas et al., 1999; Augusto et al., 2008). By improving soil fertility and moisture levels, soil amendments can increase the number and diversity of plant species that grow in the forest (Zarfes et al., 2019).

Our results also revealed that the diversity structure of understory plant communities changed from 2018 to 2019 and returned to an intermediate stage in the following years. The diversity structure differs between seasons with higher diversity in summer than in fall. This finding aligns with earlier research indicating that metrics of the recovery of plant communities show results that are impacted by seasonal measurement (Beese et al., 2022a, 2022b; Lanta et al., 2023). In early summer, as the snow melts and the temperature rises, a wide variety of plants and shrubs are competing for sunlight and resources (Lapointe, 2001; Gilliam, 2014; Dion et al., 2016). In the fall, as temperatures begin to cool and daylight hours shorten, the understory vegetation begins to prepare for winter, and some species disappear and may return in the spring (Gilliam, 2014).

4.2. Influential drivers of understory plant community diversity

The cumulative impact of soil scarification and amendment emerged as a central driver among all metrics, underscoring the profound influence of forest management practices on understory vegetation. This finding is consistent with our hypothesis and resonates with the work of Jaeger et al. (2022), who highlighted the importance of soil disturbance regimes in shaping forest understory communities. Additionally, the significance of abiotic factors, particularly light availability and soil moisture, aligns with existing literature on their roles in understory dynamics (Landuyt et al., 2019; Dormann et al., 2020). The impact of biotic factors, such as beech shrub and seedling abundance, supports studies emphasizing the influence of dominant species on understory composition (Kaushik et al., 2021). The identification of beech shrub and seedling abundance as key drivers emphasizes the lasting impact of pre-and post-site preparation conditions on understory vegetation. This echoes the findings of Perring et al. (2016), emphasizing the legacy effects of land-use history on forest dynamics.

The nuanced effects observed between soil amendment with and without scarification reflect the complex interplay between different forest management practices. Similar findings have been reported by Jaeger et al. (2022), highlighting the need for tailored approaches in forest ecosystem management to achieve desired biodiversity outcomes. The increased importance of light availability and the decreased influence of soil moisture reveal the temporal-induced shifts in understory diversity structure (Murphy and McCarthy, 2014; Hédil and Chudomelová, 2020).

The pronounced role of seasonality in influencing functional redundancy and uniqueness underscores the need to consider temporal variability in assessing forest understory dynamics. This finding complements previous studies (Beese et al., 2022a, b; Lanta et al., 2023), highlighting the significance of seasonality in shaping plant community traits.

While microsite drivers beyond the ones examined in this study showed relatively lower importance, the hierarchical dominance of abiotic factors over biotic factors aligns with the broader ecological literature emphasizing the primary role of environmental conditions in structuring plant communities (Grime, 2007; Landuyt et al., 2019). Our results emphasize the need to focus on managing abiotic factors effectively to promote and conserve understory plant diversity, especially in areas undergoing site preparation and other disturbances.

4.3. Trait assemblages' associations with microsite conditions

The integration of RLQ, fourth-corner, and CWM analyses provides a comprehensive understanding of how microsite conditions, particularly soil properties, shape the functional traits of the understory plant community. These findings support the niche theory (Ricotta et al., 2023a; Senthilnathan and D'Andrea, 2023), highlighting the role of microsite conditions in filtering species with specific traits over time. The RLQ analysis highlighted that certain soil treatments, particularly scarified soil with ash and lime amendments and non-scarified soil with fertilizer amendment, exhibited stronger associations with functional traits compared to other environmental variables. The fourth-corner method reinforced these associations, revealing significant links between microsite conditions and functional trait characteristics. The synergistic effects of scarification and lime amendment, for instance, favored specific functional traits, such as hemicryptophyte, mesic-xeric, and semi-persistent species, highlighting the role of combined treatments in shaping community composition. The varying impacts of different amendments, including positive effects on certain traits and negative effects on others, underscore the complexity of soil treatments and their consequences for functional trait dynamics. The analysis of community-weighted mean traits (CWM) further delves into the functional trait composition of the understory plant community, elucidating the profound influence of soil scarification combined with various

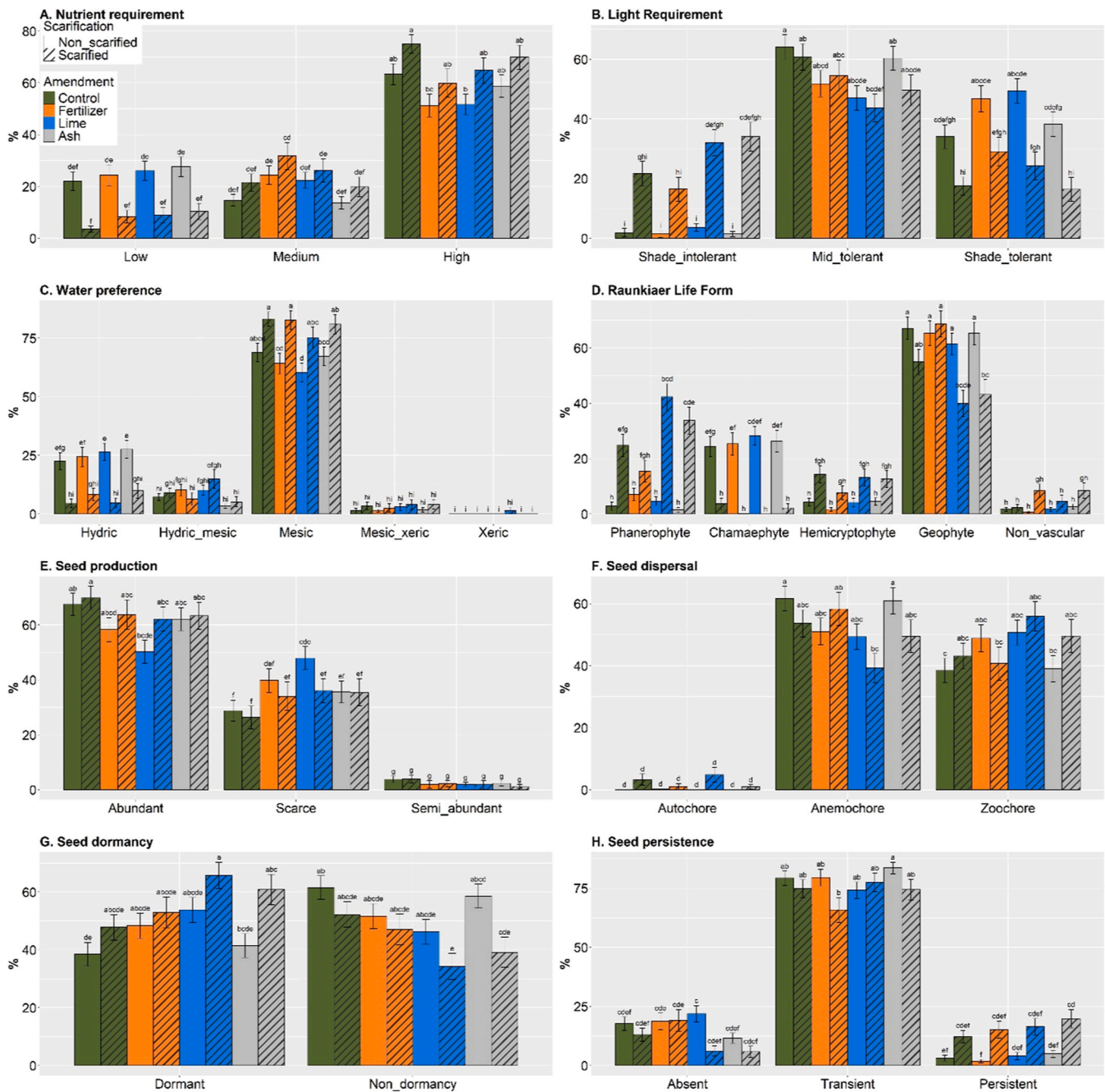


Fig. 4. Illustration of the effect of soil scarification and amendment on the trait community weighted mean (CWM $\pm$ SE). The percentages represent the proportion of trait categories within each treatment. The percentages of trait categories within a treatment should add up to 100 %.

amendments. The dominance of traits associated with high nutrient levels, moderate shade tolerance, and mesic moisture conditions indicates a shift towards a community structure favoring species with robust growth strategies under scarified and amended conditions. Conversely, the application of amendments in conjunction with scarification mitigates traits associated with lower nutrient needs, high shade tolerance, and preference for hydric conditions. This suggests a shift away from species adapted to nutrient-poor, shaded, or overly moist environments, highlighting the role of amendments in altering the plant community towards a more dynamic and nutrient-responsive assembly.

Importantly, our results highlight that environmental variables such as temporal light variation, temporal soil moisture fluctuation, and the presence of tree seedlings and saplings were more significantly associated with trait categories than specific soil treatments. This suggests that

while soil treatments play a crucial role, broader environmental conditions contribute significantly to shaping the traits and composition of the understory plant community. The observed differentiation of functional traits over time based on soil properties and light gradients suggests the adaptive nature of plant species to particular soil conditions, a key determinant influencing plant community composition and diversity.

Early succession in our study site is characterized by the dominance of high-resource exploitation traits, such as high nutrient requirements and strong colonization abilities, particularly exhibited by geophytes. This dominance aligns with ecological theories proposing that after disturbance (e.g. scarification), species with traits favoring resource exploitation and rapid colonization prevail initially (Grime, 1977). Geophytes, equipped with subterranean organs and early sprouting capabilities, demonstrate resilience in early successional environments,

where resource availability is often heightened (Tilman, 2004). Soil scarification further promotes these traits by facilitating suckering and germination.

The shift in dominance towards mid-tolerant species suggests limited light availability following a selection cut at our site. However, soil scarification, by enhancing light availability, influences the composition by decreasing the dominance of mid-tolerant species and favoring shade-intolerant ones. This supports biological interaction hypotheses indicating a progression from colonization-oriented traits to competition-oriented traits over successional stages (Grime, 1977; Webb et al., 2002).

Our study reveals a prevalence of species with transient seed and the co-dominance of anemochore and zoochore dispersal mechanisms, supporting neutral theory concepts (Webb et al., 2002; Tilman, 2004). The dominance of species with easily dispersed seeds, irrespective of their specific ecological requirements, suggests the influence of ecological drift and stochastic processes in community assembly. Soil scarification, by creating favorable microsites, facilitates the recruitment of species adapted to open and disturbed habitats, further contributing to these stochastic processes.

4.4. Study limitations

Despite the rigorous design of the study, certain limitations should be acknowledged. Firstly, the relatively small plot size of 8×4 m, with subplots of 2×2 m, while larger than the typical 1 m^2 plots used for assessing understory vegetation (Gayton, 2014; Bolton and D'Amato, 2019) may not fully capture the spatial heterogeneity typical of forest ecosystems. This spatial constraint can limit the generalizability of our findings, as ecological processes observed in smaller plots might not reflect those in larger, more diverse areas. Additionally, having only 12 main plots reduces the statistical power of our study, making it more challenging to detect significant treatment effects and increasing the risk of Type II errors.

These limitations may impact the robustness and broader applicability of our results. The limited number of plots and subplots means our study may not adequately represent the complexity and variability of forest understory communities. To address these issues in future research, we recommend increasing the number of plots and expanding plot sizes to enhance statistical power and ecological relevance. Moreover, conducting multi-site studies across different forest types and extending the duration of the experiment to observe long-term effects would provide a more comprehensive understanding of the impact of soil treatments on understory vegetation. These strategies would help mitigate the limitations of our current study and contribute to more robust and generalizable findings in forest ecology research.

5. Conclusion

This study sheds light on the complex relationships between soil treatments, environmental conditions, and functional traits, providing valuable insights into the mechanisms shaping the understory plant community in northern hardwood forests. Specifically, the combination of soil scarification with amendments, including nitrogen fertilizer, lime, or wood ash, induced significant changes in the diversity structure of the understory plant communities. Notably, while soil scarification, when coupled with amendments, led to a reduction in diversity, lime amendment in the absence of scarification resulted in increased diversity. This alteration in diversity structure carries both negative and positive implications for the stability and resilience of the community. Abiotic factors, particularly soil moisture and light availability, emerged as pivotal drivers shaping the plant functional structure, overshadowing the comparatively minor influence of biotic factors. These findings emphasize the need for caution among land managers considering the implementation of soil scarification and amendments, especially in combination. Despite their positive effects on the regeneration and

growth of high-value tree species like sugar maple and yellow birch, these treatments may have unintended consequences on the productivity, diversity, and resilience of the overall northern hardwood forest understory community. To reconcile the dual objectives of preserving plant diversity and enhancing the growth of high-value tree species (sugar maple, yellow birch), a judicious approach is recommended. Implementing scarification and amendment on small, well-distributed patches, rather than applying these treatments extensively across continuous areas, could be a viable strategy. This targeted application may help mitigate the temporary alterations in the stability and resilience of the understory plant community, thereby fostering a balanced and sustainable management approach for northern hardwood forests. Land managers are encouraged to consider this nuanced strategy to ensure the long-term health and biodiversity of these ecologically significant ecosystems.

CRediT authorship contribution statement

David Paré: Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Fidèle Bognounou:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Jérôme Laganière:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used chatgpt in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was funded by the Canadian Forest Service through the collaborative research project 3.3, “Sustainable supply of wood fibre for bio-economy opportunities,” hosted by the Canadian Wood Fibre Centre. Their generous contribution made this research possible and greatly enhanced our understanding in this field. We thank the field and lab crew members, including S. Dagnault, F. Michaud, O. Jeffrey, A. Cotton-Gagnon, M. Nolan, I. Nicolae, M-S. Morneau, M. Létourneau, E. Pouliot, J-M Béland, O. Couture and J-F Paquet for helping with plot establishment and monitoring as well as soil analysis. We are grateful to Bernard Ferland from Resolute Forest Products for supplying wood ash. We thank D. Correia for his comments on earlier version of this manuscript.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.foreco.2024.122089](https://doi.org/10.1016/j.foreco.2024.122089).

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